Related Research: Paddy Land Watch

Oct 24, 2026 · 10 related papers in simple words

Start here

Each summary says what the paper tried, what it found, whether it worked, and why it matters for our project. The papers follow the order our project works: paddy loss in Sri Lanka, mapping paddy, deep learning with radar, then ready-made global maps to compare against.

- **PDFs:** in the **Papers** folder next to these notes. Each file name starts with the paper's number (01 to 10).
- **Code:** in the **GitHub** folder next to these notes.
- **Only time for three?** Read papers **4**, **6** and **1**. Together they cover Sri Lankan rice maps, radar deep learning in a two-season tropical climate, and paddy loss in Gampaha.

The 10 papers at a glance

None of the five Sri Lankan studies uses Sentinel-1 radar; the radar deep-learning work here comes from China and India.

#	Paper	What it gives us	Worked?
1	Weerasinghe et al. 2025: paddy loss in Gampaha	Background: Gampaha lost most of its paddy in ten years (preprint)	Partly
2	Gunatilake et al. 2026: wetland loss in Colombo	How fast wet land next to our districts is being built over	Yes
3	Jayasooriya et al. 2023: the Southern Expressway and paddy	Paddy loss gathers near new roads, a clue for ranking alerts	Partly
4	Özdoğan et al. 2025: rice maps for every season in Sri Lanka	Ready rice maps for 2010–2024 to mark land that used to be paddy	Yes
5	Sankavi et al. 2025: CNN paddy detection in the North Central Province	The closest Sri Lankan deep-learning paddy study	Partly

#	Paper	What it gives us	Worked?
6	Shen & Liao 2025: Sentinel-1 rice maps on Hainan Island	Radar time series with LSTM and U-Net in a two-season tropical climate	Yes
7	Onojeghuo et al. 2023: ResU-Net with Sentinel-1 and Sentinel-2	Radar plus optical works best; how to make paddy change maps	Yes
8	Kasture et al. 2025: abandoned paddy from radar	Active versus abandoned paddy with a BiLSTM, like our idle class	Yes, abstract only
9	Brown et al. 2022: Dynamic World	A free built-area signal and a baseline to beat	Yes
10	Brown et al. 2025: AlphaEarth Foundations	Free yearly embeddings for a quick baseline classifier	Yes

A. Paddy loss in Sri Lanka

Paddy and wet land near Colombo are being filled and built on. These studies measure the loss, but with 30 m Landsat images and weak accuracy checks.

1. Paddy land loss in Gampaha (Weerasinghe et al., 2025)

Weerasinghe, V. P. A., Vidanagama, U. S., & Senadeera, T. S. M. (2025). Mapping paddy lands using remote sensing techniques and land use land cover change detection over a decade in Gampaha District, Sri Lanka [Preprint]. *Research Square*. Open the paper

Problem. Gampaha is growing fast and losing paddy land. The authors compared simple ways of mapping paddy and measured ten years of change.

What they did. They used Landsat 5 (2008/09) and Landsat 8 (2018/19) images from three dates in the Maha season. They compared three methods in ArcGIS: supervised classification, Tasselled Cap with NDVI, and LSWI with NDVI (LSWI above NDVI means a flooded field). Results were checked against Google Earth points.

What they found. Each method worked best at a different crop stage, reaching 84–90% accuracy at its best stage (LSWI with NDVI fell to 53% in 2018/19). Paddy fell from 283.81 km² in 2009 to 66.74 km² in 2019, a 76.5% drop. Built-up land grew by 126.4%.

Did it work? Partly. The methods found paddy, but the 2019 land-use map was only 70.4% accurate.

Weak points. It is a preprint, not peer reviewed, and 30 m pixels are coarse for small fields. Wetlands may be mixed up with paddy.

Why it matters for us. It covers our own district. Of the 2009 paddy, only about 25% became built-up land, while 53% became forest or other vegetation. The weak 2019 map shows why we need radar that sees through cloud and 10 m data. Treat the 76.5% figure with care.

Short on time? Read the results (the accuracy table and the 2009–2019 change) and the discussion section on changes in paddy fields.

2. Wetland loss in Colombo (Gunatilake et al., 2026)

Gunatilake, U., Weerasinghe, V. P. A., & Wickramaratne, C. (2026). Wetland loss, impervious surface expansion, and urban thermal stress: A spatiotemporal analysis of land use change and urban thermal patterns in Colombo District, Sri Lanka. *Biosphere*, 2(3), 8. Open the paper

Problem. Colombo is losing wetlands to buildings and roads, and it is getting hotter. The authors asked whether land cover affects heat the same way across the district.

What they did. They used Landsat 5, 7, 8 and 9 in Google Earth Engine for seven years between 1989 and 2024. A Random Forest mapped four classes: wetland plants, water, built surface and other open land. They then linked land surface temperature to land cover on a 500 m grid.

What they found. Wetland plant cover dropped from about 200 km² in 1989 to 52 km² in 2024, a loss of about 74%. Built surface grew from 42 km² to 179 km² (+326%). The 2024 map was 99.5% accurate against independent ground truth.

Did it work? Yes. The change maps are clear, though the older years were not checked independently.

Weak points. Pixels are 30 m, and for 1989–2019 the accuracy only shows agreement with threshold masks. There is no separate paddy class.

Why it matters for us. Colombo borders Gampaha and Kalutara, and this shows how fast nearby wet land is being built over. It is a Random Forest baseline; we separate paddy from other wetland and work season by season.

Short on time? Read Section 2 (methods, pp. 3–7), Section 3.1 (land-use change) and Section 4.7 (limitations).

3. The Southern Expressway and paddy loss (Jayasooriya et al., 2023)

Jayasooriya, J. A. H. D., Dinusha, K. A., Gunasinghe, G. P., & Ranasinghe, A. K. R. N. (2023). Monitoring the impact on paddy fields during the construction of the Southern Expressway using remote sensing and GIS. *KDU International Research Conference (IRC 2023)*, 8–13. Open the paper

Problem. The Southern Expressway may be causing paddy loss nearby in Hambantota. Field surveys to check this are slow and costly.

What they did. They classified Landsat images (30 m) from 1990 to 2022 into six classes in ArcGIS. They drew a 2 km zone on each side of the expressway and cut it into 500 m strips. Then they tested whether paddy area changes with distance from the road.

What they found. Paddy inside the 2 km zone fell from 107 km² in 2015 to 99 km² in 2019, then rose to 101 km² in 2022. District paddy hardly changed (about 595 km² in 1990 and 604 km² in 2020). There was more paddy further from the road (correlations 0.82–0.97).

Did it work? Partly. It shows paddy loss close to the road, but gives no map accuracy figures.

Weak points. No overall accuracy or kappa, and Google Earth was used instead of field checks. Clouds forced 5-year gaps before 2015.

Why it matters for us. Paddy loss in Sri Lanka gathers near new roads, so distance to new roads could help rank our alerts. We must report the accuracy this paper leaves out.

Short on time? Read the methodology (pp. 9–10) and the results (pp. 10–12).

B. Mapping paddy in Sri Lanka

Two recent studies map Sri Lankan paddy well with optical images. Neither uses radar, and neither looks for filled land.

4. Rice maps for every season (Özdoğan et al., 2025)

Özdoğan, M., Wang, S., Ghose, D., Fraga, E., Fernandes, A., & Varela, G. (2025). Field-scale rice area and yield mapping in Sri Lanka with optical remote sensing and limited training data. *Remote Sensing*, 17(17), 3065. Open the paper

Problem. Government paddy figures exist only for whole districts, so nobody sees field-by-field change. Clouds also make satellite mapping hard in Sri Lanka.

What they did. They used Landsat (2000–2022) and Sentinel-2 (from 2019) in Google Earth Engine. A simple rule on NDVI and NDWI marked rice pixels for every Maha and Yala season, so very few training labels were needed. A random forest then predicted yield.

What they found. The rice maps were about 91% accurate for Maha 2021 and 93% for Yala 2021; across all checked years the paper gives 70% to over 90%. The yield model had R^2 0.65 and an error of 451 kg/ha (12%). Mapped areas followed government figures closely, but less well in small districts.

Did it work? Yes for rice maps with few labels; the yield estimates were only fair.

Weak points. The maps use 30 m pixels, not real field boundaries, and yield was not checked in the field. Accuracy was tested only for recent years.

Why it matters for us. Their ready Maha and Yala rice maps for 2010–2024 give us a "this used to be paddy" layer and help choose training points in Gampaha and Kalutara. We go further with Sentinel-1 radar, deep learning, four classes and filling alerts.

Short on time? Read Section 2.3 (method, pp. 6–9), Section 3.1 (accuracy, pp. 13–14) and Section 4 (pp. 20–22).

5. CNN paddy detection in the North Central Province (Sankavi et al., 2025)

Sankavi, K., Madusanka, N. B. S., & Jayasinghe, A. (2025). Integrating CNN deep learning and biomass correlation approaches for paddy land detection and yield estimation in North Central Province, Sri Lanka. *FARU Proceedings 2025*, University of Moratuwa. Open the paper

Problem. Paddy land and harvest figures are often late or wrong, which leads to poor decisions on rice imports.

What they did. They used Sentinel-2 images of Anuradhapura and Polonnaruwa for Maha and Yala seasons from 2017/18 to 2023/24. A U-Net found paddy land after training on 5,000 paddy and 5,000 non-paddy samples. NDVI and NDWI cut-offs split active from inactive fields, and a formula estimated yield before harvest.

What they found. The CNN had 88.5% overall accuracy, and 93.5–98.5% for single seasons. But the area estimate was 9% off in Maha 2017/18 and 37% off in Maha 2023/24.

Did it work? Partly. The CNN found paddy well, but the area and yield figures swung a lot between seasons.

Weak points. Results were compared only with government figures, with no field checks. No precision, recall or F1 is reported.

Why it matters for us. It is the closest Sri Lankan deep-learning paddy study, and its active versus inactive split is like our paddy versus idle classes. We add Sentinel-1 radar for the cloudy wet-zone seasons and a filled-or-built class.

Short on time? Read Section 3.1 (finding active paddy), Section 4.1 (results tables) and Section 5.3 (limitations).

C. Deep learning with radar

Radar sees through monsoon cloud, which is why these three studies matter most for the wet zone. All three come from China or India.

6. Sentinel-1 rice maps on Hainan Island (Shen & Liao, 2025)

Shen, G., & Liao, J. (2025). Paddy rice mapping in Hainan Island using time-series Sentinel-1 SAR data and deep learning. *Remote Sensing*, 17(6), 1033. Open the paper

Problem. In rainy tropical places, clouds block optical images. Can rice be mapped in all weather with Sentinel-1 radar alone?

What they did. They used Sentinel-1 VV and VH time series for 2020 and 2023 over Hainan Island, China, which has an early and a late rice season. They built three models: an LSTM that reads each pixel's time series, a U-Net that reads the image stack, and a model joining both. Labels came from 490 paddy polygons.

What they found. For late rice in 2020, the joined model reached accuracy 0.98 and F1 0.91. Adding attention helped the LSTM. Early-rice area was about 2% off official figures, but late-rice area was about 30% too low.

Did it work? Yes per pixel, but the total area for late rice was off.

Weak points. Roads and field ridges were often labelled as rice, and radar speckle made the maps noisy. The model is heavy, needing about 48 GB of GPU memory.

Why it matters for us. It is the closest match to our radar part: tropical, two seasons, Sentinel-1 time series, LSTM plus U-Net. We need a lighter model and more classes (other crop, idle, filled).

Short on time? Read Sections 3.2–3.4 (models and samples), Table 1 in Section 4.2 and Section 5.

7. Radar plus optical with a ResU-Net (Onojeghuo et al., 2023)

Onojeghuo, A. O., Miao, Y., & Blackburn, G. A. (2023). Deep ResU-Net convolutional neural networks segmentation for smallholder paddy rice mapping using Sentinel 1 SAR and Sentinel 2 optical imagery. *Remote Sensing*, 15(6), 1517. Open the paper

Problem. Small, oddly shaped paddy plots under frequent cloud are hard to map. The authors wanted a practical method with free Sentinel data.

What they did. They mapped five districts in Heilongjiang, China, for 2016, 2018 and 2020. They tested seven input sets mixing Sentinel-1 (one image per growth stage) and Sentinel-2 (one median image for the season). They compared a ResU-Net with Random Forest at 1,000 random check points.

What they found. Sentinel-2 bands plus Sentinel-1 was the best input. ResU-Net beat Random Forest: overall accuracy 94% against 91%, and mean F1 0.92 against 0.89. Their change maps showed 437,123 ha of paddy becoming other land between 2016 and 2018.

Did it work? Yes. Radar plus optical with a U-Net clearly beat Random Forest.

Weak points. The deep model needed many training labels, and only one region was tested. Some figures in the abstract, tables and conclusion do not match.

Why it matters for us. It shows that adding Sentinel-1 to Sentinel-2 helps a U-Net map small paddy fields. It also shows how yearly maps become "paddy lost" change maps, which is the base of our filling alerts.

Short on time? Read Section 2.2 (workflow), Sections 3.1–3.2 (best inputs; U-Net against Random Forest) and Section 3.4 (change from 2016 to 2020).

8. Abandoned paddy from radar (Kasture et al., 2025)

Kasture, S., Hegde A, A., & Umesh, P. (2025). Deep learning based paddy land abandonment detection using multitemporal polarimetric SAR patterns. *EGU General Assembly 2025*, EGU25-18606. Open the abstract

Problem. Many paddy fields in Udupi district (coastal Karnataka, India) are being abandoned, and clouds block optical images there.

What they did. They built a time series of Sentinel-1 VV and VH radar values for each pixel over the main (Kharif) season. They added three polarimetric features: entropy, anisotropy and alpha. LSTM, BiLSTM and CNN-BiLSTM models were trained on fields known to be active or abandoned.

What they found. BiLSTM was best, with 94% overall accuracy and kappa 0.907. It also had better precision and recall for abandoned fields.

Did it work? Yes, but this is only an abstract and slides, not a reviewed paper.

Weak points. Not peer reviewed, one district only, and few details on sample sizes. The abstract says 2017–2024, but the slides say 2021–2024.

Why it matters for us. It is very close to our idle class: radar time series with a BiLSTM separate active from abandoned paddy in a rainy region. We can start with VV and VH and test polarimetric features later.

Short on time? Read the abstract page, then the data and results slides. The slides may need a free EGU login.

D. Ready-made global maps

These free global products give quick baselines. Our model must beat them on paddy, idle and filled land.

9. Dynamic World (Brown et al., 2022)

Brown, C. F., Brumby, S. P., Guzder-Williams, B., Birch, T., Hyde, S. B., Mazzariello, J., Czerwinski, W., Pasquarella, V. J., Haertel, R., Ilyushchenko, S., Schwehr, K., Weisse, M., Stolle, F., Hanson, C., Guinan, O., Moore, R., & Tait, A. M. (2022). Dynamic World, near real-time global 10 m land use land cover mapping. *Scientific Data*, 9, 251. Open the paper

Problem. Normal land-cover maps come once a year, and late. The authors wanted a global 10 m map that updates with every new Sentinel-2 image.

What they did. People labelled about 24,000 Sentinel-2 tiles. A small neural network gives each pixel a probability for 9 classes, including crops, built area and flooded vegetation. It runs in Earth Engine on about 12,000 scenes a day.

What they found. It agreed with expert labels 73.8% of the time, beating most other products tested (54.8–67.4%). Built area was good (about 88% recall), crops weaker (about 60%), and flooded vegetation (about 37%) and grass (about 34%) weakest.

Did it work? Yes as a live map, but it is weak for crops and flooded vegetation.

Weak points. It uses optical images only, so missed clouds and shadows cause wrong labels. Single-date maps are noisy, and there is no paddy or idle class.

Why it matters for us. Its built-area probability is a free extra signal that a paddy tract has been filled or built on, and a baseline to beat. Its weak crop and flooded classes show why we need our own radar plus optical paddy model.

Short on time? Read Methods (model training and near real-time inference), Technical Validation and Usage Notes.

10. AlphaEarth Foundations (Brown et al., 2025)

Brown, C. F., Kazmierski, M. R., Pasquarella, V. J., Rucklidge, W. J., Samsikova, M., Zhang, C., Shelhamer, E., Lahera, E., Wiles, O., Ilyushchenko, S., et al. (2025). AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data. arXiv:2507.22291. Open the paper

Problem. There is a huge amount of satellite data but very few labels. The authors wanted one general summary for every 10 m pixel, so that mapping works with only a few labels.

What they did. One model learned from many sources, including Sentinel-2, Landsat, Sentinel-1, elevation and climate data. It gives every 10 m pixel 64 numbers (an embedding) for each year from 2017 to 2024. They tested it on 15 mapping tasks with simple classifiers.

What they found. With the most labels they tested (tens to hundreds per task), errors were about 23.9% lower on average than the next-best method; with 10 labels per class, about 10.4% lower. Gains were largest for yearly maps such as crop masks and land use.

Did it work? Yes, best overall, but not on every task.

Weak points. The maps show tile artifacts. The released layers are yearly, so they cannot separate Yala from Maha.

Why it matters for us. The embeddings are free in Earth Engine as Satellite Embedding V1, so we can train a quick classifier on them as a baseline. We then compare it with our seasonal Sentinel-1 plus Sentinel-2 model.

Short on time? Read the introduction and the sections on data-scarce evaluation, change detection and the global embeddings dataset.

GitHub code in the GitHub folder

The Özdoğan repository links ready rice maps of Sri Lanka, so it is the most useful of the three. All three are in the **GitHub** folder; open each README first.

Folder	What it is	Licence	How we would use it
ozdogan15-srilanka	Code and data for paper 4: Earth Engine scripts, ready Maha and Yala rice maps for 2010–2024 (30 m) as public assets, and yield features	None given	Clip the rice maps (`projects/ee-ozdogan05/assets/srilanka/LK_maha_rice_v3` and `LK_yala_rice_v3`) to Gampaha and Kalutara as a historic-paddy mask. Use it to pick training points and spot paddy that stops being paddy. Ask the author before reusing the code.

Folder	What it is	Licence	How we would use it
WorldCereal-classification	ESA WorldCereal's Python package for cropland and crop-type maps, run in the cloud with openEO and built on the Presto pretrained transformer	MIT	Make a crop or no-crop layer for one Yala or Maha season to compare with ours. Later, train a paddy model on top of Presto. Free use is capped by monthly CDSE credits.
dl4sits-BreizhCrops	A French crop-type time-series dataset plus PyTorch models: LSTM, TempCNN, Transformer, InceptionTime and more	GPL-3.0	Reuse its time-series models as baselines on our Sentinel-1 and Sentinel-2 sequences for each tract. Shared code that includes theirs must also be GPL.

What these papers tell us

None of the five Sri Lankan studies uses Sentinel-1 radar, and none watches paddy being filled season by season. The radar deep-learning methods exist, but they were built for China and India.

- **The loss is real and fast.** Gampaha's paddy fell about 76% in a decade (paper 1, a preprint), and nearby Colombo lost about 74% of its wetland plants since 1989 (paper 2).
- **Sri Lankan maps use optical images.** Rice maps reach 91–93% accuracy (paper 4) and a U-Net reached 88.5% (paper 5), but cloud limits both in the wet zone.
- **Radar deep learning works in two-season tropics.** An LSTM plus U-Net on Sentinel-1 reached F1 0.91 (paper 6), and radar plus optical was the best input (paper 7).
- **Idle land can be detected.** A BiLSTM on radar told active from abandoned paddy with 94% accuracy (paper 8, abstract only).

- **Global maps give baselines, not answers.** Dynamic World is weak on crops (paper 9), and AlphaEarth is yearly, not seasonal (paper 10).
- **Accuracy checks are often thin.** Paper 3 reports no map accuracy, and paper 5 checked areas only against government figures.

The gap our project fills: a Yala and Maha map of every paddy tract in Gampaha and Kalutara as paddy, other crop, idle or filled, made from Sentinel-1 and Sentinel-2 with deep learning, with ranked filling alerts for the Department of Agrarian Development.

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APA style, ready to copy into the report.

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